ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 16	Even number of values: mean of the two middle values
2	**(d)** 20	Total − known sum
3	**(c)** 22	New sum 132, new count 6
4	**(a)** Between week 2 and week 3	Growth in each week: 4, 7, 4, 3, 6; largest is 7
5	**(d)** 7	The value with most dots is 7
6	**(c)** 6	$\sum xf = 72$, $\sum f = 12$
7	**(a)** 19	Even number of values: mean of the two middle values
8	**(c)** 26	Total − known sum
9	**(d)** 30	New sum 180, new count 6
10	**(b)** Between week 3 and week 4	Growth in each week: 3, 2, 6, 0, 0; largest is 6

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ascending: 12, 21, 34, 38, 44, 45 (1); mean of the two middle values $= 36$ (1).	1+1
12	Total $= 4 \times 27 = 108$ (1); fourth $= 108 - 76 = 32$ (1).	1+1
13	Old mean $= 24$ (1); new mean $= 28$ (adds 4) (1).	1+1
14	Ascending: 9, 10, 19, 35, 42, 60 (1); mean of the two middle values $= 27$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mean $= 61$ (1); median $= 66.5$ (1); range $= 58$ (1).	1+1+1
16	$\sum xf = 51$ (1); $\sum f = 10$ (1); mean $= 5.1$ (1).	1+1+1
17	Sum $= 120$ (1); new sum $= 108$ over 4 values (1); mean $= 27$ (1).	1+1+1
	OR: Mean $= 48$ (1); median $= 36$ (1); range $= 74$ (1).	1+1+1
18	$\sum xf = 240$ (1); $\sum f = 10$ (1); mean $= 24$ (1).	1+1+1
	OR: Sum $= 280$ (1); new sum $= 210$ over 6 values (1); mean $= 35$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) mean $= 130$ (1); (ii) rises -98, -39, 129, -25, 3; greatest in month September (2); (iii) range $= 137$ (2).	1+2+2
	OR: Mean A $= 41$, mean B $= 37$ (2); range A $= 62$, range B $= 29$ (2); more consistent: B (1).	2+2+1
20	(i) mean $= 121$ (1); (ii) rises -100, -9, 90, 27, 20; greatest in month September (2); (iii) range $= 137$ (2).	1+2+2
	OR: Mean A $= 45$, mean B $= 64$ (2); range A $= 42$, range B $= 45$ (2); more consistent: A (1).	2+2+1

Mistakes to avoid:

- Finding the median without arranging the data in order first.
- For an even number of observations, taking one of the middle values instead of the **mean of the two middle values**.
- Confusing mode with median or mean: mode is the most frequent value, not the middle value.
- Dividing the sum by the number of *distinct* values instead of the total number of observations (counting repeated values only once).
- Computing range as the difference between the first and last values as listed, not the maximum and minimum.
- Reading a line graph at the wrong point or ignoring the scale of the axes.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 3	The value with most dots is 3
2	**(d)** 6	$\sum xf = 84$, $\sum f = 14$
3	**(b)** 37	Even number of values: mean of the two middle values
4	**(a)** 35	Total − known sum
5	**(b)** 30	New sum 180, new count 6
6	**(a)** Between week 3 and week 4	Growth in each week: 3, -1, 5, -2, 2; largest is 5
7	**(c)** 6	The value with most dots is 6
8	**(a)** 5	$\sum xf = 100$, $\sum f = 20$
9	**(b)** 28	Even number of values: mean of the two middle values
10	**(d)** 20	Total − known sum

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total $= 4 \times 36 = 144$ (1); fourth $= 144 - 102 = 42$ (1).	1+1
12	Old mean $= 19$ (1); new mean $= 26$ (adds 7) (1).	1+1
13	Ascending: 28, 29, 34, 37, 42, 56 (1); mean of the two middle values $= 35.5$ (1).	1+1
14	Total $= 4 \times 19 = 76$ (1); fourth $= 76 - 55 = 21$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mean $= 54$ (1); median $= 48$ (1); range $= 60$ (1).	1+1+1
16	$\sum xf = 36$ (1); $\sum f = 10$ (1); mean $= 3.6$ (1).	1+1+1
17	Sum $= 120$ (1); new sum $= 80$ over 4 values (1); mean $= 20$ (1).	1+1+1
	OR: Mean $= 40$ (1); median $= 36$ (1); range $= 55$ (1).	1+1+1
18	$\sum xf = 320$ (1); $\sum f = 10$ (1); mean $= 32$ (1).	1+1+1
	OR: Sum $= 245$ (1); new sum $= 210$ over 6 values (1); mean $= 35$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) mean $= 85$ (1); (ii) rises -93, -39, -13, -7, 62; greatest in month November (2); (iii) range $= 152$ (2).	1+2+2
	OR: Mean A $= 41$, mean B $= 42$ (2); range A $= 28$, range B $= 68$ (2); more consistent: A (1).	2+2+1
20	(i) mean $= 94$ (1); (ii) rises 53, -12, -26, 121, 15; greatest in month October (2); (iii) range $= 151$ (2).	1+2+2
	OR: Mean A $= 53$, mean B $= 52$ (2); range A $= 41$, range B $= 51$ (2); more consistent: A (1).	2+2+1

Mistakes to avoid:

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 26	New sum 156, new count 6
2	**(b)** Between week 3 and week 4	Growth in each week: 0, 4, 5, 0, -2; largest is 5
3	**(c)** 3	The value with most dots is 3
4	**(d)** 9	$\sum xf = 108$, $\sum f = 12$
5	**(c)** 28	Even number of values: mean of the two middle values
6	**(a)** 52	Total − known sum
7	**(d)** 22	New sum 132, new count 6
8	**(b)** Between week 2 and week 3	Growth in each week: -3, 6, 3, 5, 4; largest is 6
9	**(a)** 5	The value with most dots is 5
10	**(d)** 8	$\sum xf = 48$, $\sum f = 6$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Old mean $= 15$ (1); new mean $= 25$ (adds 10) (1).	1+1
12	Ascending: 16, 24, 32, 46, 54, 57 (1); mean of the two middle values $= 39$ (1).	1+1
13	Total $= 4 \times 52 = 208$ (1); fourth $= 208 - 153 = 55$ (1).	1+1
14	Old mean $= 18$ (1); new mean $= 21$ (adds 3) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mean $= 37$ (1); median $= 34.5$ (1); range $= 47$ (1).	1+1+1
16	$\sum xf = 69$ (1); $\sum f = 10$ (1); mean $= 6.9$ (1).	1+1+1
17	Sum $= 75$ (1); new sum $= 60$ over 4 values (1); mean $= 15$ (1).	1+1+1
	OR: Mean $= 54$ (1); median $= 52.5$ (1); range $= 67$ (1).	1+1+1
18	$\sum xf = 610$ (1); $\sum f = 10$ (1); mean $= 61$ (1).	1+1+1
	OR: Sum $= 252$ (1); new sum $= 216$ over 6 values (1); mean $= 36$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) mean $= 146$ (1); (ii) rises -73, 51, -31, 74, -62; greatest in month October (2); (iii) range $= 94$ (2).	1+2+2
	OR: Mean A $= 52$, mean B $= 59$ (2); range A $= 62$, range B $= 48$ (2); more consistent: B (1).	2+2+1
20	(i) mean $= 148$ (1); (ii) rises 101, -2, -45, 61, 8; greatest in month July (2); (iii) range $= 123$ (2).	1+2+2
	OR: Mean A $= 36$, mean B $= 47$ (2); range A $= 61$, range B $= 75$ (2); more consistent: A (1).	2+2+1

Mistakes to avoid:

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